Question Bank and PYQs

2 Mark Questions

Q1: Define the term 'Resource' based on the NCERT curriculum. (2 Marks)

- Everything available in our environment which can be used to satisfy our needs is a resource .
- It must be technologically accessible .
- It must be economically feasible .
- It must be culturally acceptable .

Q2: Classify resources on the basis of origin. (2 Marks)

- Biotic Resources: These are obtained from the biosphere and have life, such as human beings, flora, and fauna .
- Abiotic Resources: These are composed of non-living things, such as rocks and metals .

Q3: What is meant by 'Sustainable Economic Development'? (2 Marks)

- Development should take place without damaging the environment .
- Development in the present should not compromise with the needs of the future generations .

Q4: State the main objective of Agenda 21. (2 Marks)

- It aims at achieving global sustainable development .
- It is an agenda to combat environmental damage, poverty, and disease through global co-operation .
- A major objective is that every local government should draw its own local Agenda 21 .

Q6: What are 'Shelter Belts' and how do they help? (2 Marks)

- Shelter belts are rows of trees planted to create shelter .
- They contribute significantly to the stabilisation of sand dunes .

- They help in stabilising the desert in western India .

Q8: What are 'ravines' and where are they commonly found in India? (2 Marks)

- Ravines are bad lands created by gully erosion where the land becomes unfit for cultivation .
- They are commonly found in the Chambal basin .

Q9: Mention two factors responsible for the formation of black soil. (2 Marks)

- Climatic conditions .
- The parent rock material (specifically lava flows from the Deccan trap).

Q10: What is the 'Stock' category of resources? (2 Marks)

- Materials in the environment which have the potential to satisfy human needs but humans do not have the appropriate technology to access them .
- For example, water is a compound of two inflammable gases, hydrogen and oxygen, which could be used as a rich source of energy, but we do not have the technical 'know-how' to use them for this purpose .

3 Mark Questions

Q5: Distinguish between Bangar and Khadar alluvial soils. (3 Marks)

- Bangar is old alluvial soil, while Khadar is new alluvial soil .
- Bangar soil has a higher concentration of 'kanker' nodules .
- Khadar has more fine particles and is more fertile than the Bangar .

Q7: Define 'Net Sown Area' and 'Gross Cropped Area'. (3 Marks)

- Net Sown Area (NSA) is the physical extent of land on which crops are sown and harvested .
- Gross Cropped Area is the area sown more than once in an agricultural year plus net sown area .

Q11: Explain the interdependent relationship between nature, technology, and institutions. (3 Marks)

- The process of transformation of things available in our environment involves an interactive relationship between nature, technology, and institutions .
- Human beings interact with nature through technology .
- They create institutions to accelerate their economic development .
- This indicates that resources are a function of human activities and humans themselves are essential components of resources .

Q12: Classify resources on the basis of ownership with examples. (3 Marks)

- Individual Resources: Owned privately by individuals, e.g., plots, houses, or plantation land .
- Community Owned Resources: Accessible to all members of the community, e.g., grazing grounds, burial grounds, or public parks .
- National Resources: Technically, all resources within the political boundaries and oceanic area up to 12 nautical miles from the coast belong to the nation .
- International Resources: Regulated by international institutions; areas beyond 200 nautical miles of the Exclusive Economic Zone belong to the open ocean .

Q13: What are the major problems caused by the indiscriminate use of resources? (3 Marks)

- Depletion of resources to satisfy the greed of a few individuals .
- Accumulation of resources in a few hands, dividing society into 'haves' and 'have-nots' or rich and poor .
- Global ecological crises such as global warming, ozone layer depletion, environmental pollution, and land degradation .

Q14: Explain the importance of resource planning in a country like India. (3 Marks)

- India has enormous diversity in the availability of resources; some regions are rich in certain resources but deficient in others .
- Jharkhand and Chhattisgarh are rich in minerals and coal but may lack other facilities .
- Arunachal Pradesh has abundant water but lacks infrastructural development .
- Rajasthan has solar and wind energy but lacks water resources .

- Balanced resource planning is necessary at national, state, regional, and local levels to address these disparities .

Q15: Enumerate the three stages of resource planning in India. (3 Marks)

- Identification and inventory: Involves surveying, mapping, and qualitative/quantitative estimation and measurement of resources .
- Evolving a planning structure: Endowed with appropriate technology, skill, and institutional setup for implementing plans .
- Matching development plans: Aligning resource development plans with overall national development plans .

Q16: Which physical and human factors determine the land use pattern in India? (3 Marks)

- Physical factors include topography, climate, and soil types .
- Human factors include population density and technological capability .
- Culture and traditions also play a role in determining how land is used .

Q17: Identify human activities that have contributed significantly to land degradation. (3 Marks)

- Deforestation, often due to mining and other industrial activities .
- Overgrazing in states like Gujarat, Rajasthan, and Maharashtra .
- Over-irrigation in Punjab and Haryana, leading to waterlogging and increased salinity .
- Mining and quarrying, which leave deep scars and traces of over-burdening on the land .

Q18: Describe the characteristics of Arid soils. (3 Marks)

- They range from red to brown in colour and are generally sandy in texture and saline in nature .
- Due to dry climate and high temperature, they lack humus and moisture .
- The lower horizons are occupied by 'kanker' due to increasing calcium content downwards .
- With proper irrigation, these soils can become cultivable, as seen in western Rajasthan .

Q19: Suggest three methods to solve the problems of land degradation in industrial and arid areas. (3 Marks)

- Planting of shelter belts and stabilisation of sand dunes by growing thorny bushes in arid areas .
- Proper management of waste lands and control of mining activities .
- Proper discharge and disposal of industrial effluents and wastes after treatment in industrial and suburban areas .

Q20: Why did Gandhiji voice his concern about resource conservation? (3 Marks)

- He believed 'There is enough for everybody's need and not for any body's greed'.
- He identified greedy and selfish individuals and the exploitative nature of modern technology as root causes of depletion.
- He was against mass production and wanted to replace it with production by the masses.

5 Mark Questions

Q21: Classify resources based on the status of development with detailed explanations. (5 Marks)

- Potential Resources: Resources found in a region but not yet utilised, such as solar and wind energy potential in Rajasthan and Gujarat .
- Developed Resources: Resources that are surveyed and their quality and quantity are determined for utilisation .
- Stock: Materials in the environment with potential to satisfy needs, but lacking the technology to be accessed, such as hydrogen from water .
- Reserves: A subset of 'stock' that can be put into use with existing technical 'know-how' but their use has not been started yet; they are kept for meeting future requirements .
- Mention a diagram classifying resources to clarify these categories .

Q22: Discuss the significance of the 1992 Rio de Janeiro Earth Summit and Agenda 21. (5 Marks)

- In June 1992, more than 100 heads of states met in Brazil for the first International Earth Summit .
- The summit addressed urgent problems of environmental protection and socio-economic development at the global level .
- Leaders signed the Declaration on Global Climatic Change and Biological Diversity .
- The convention adopted Agenda 21 for achieving Sustainable Development in the 21st century .
- Agenda 21 aims to combat environmental damage, poverty, and disease through global co-operation on common interests .
- It emphasizes that every local government should create its own local version of Agenda 21 .

Q23: Describe the three major relief features of India and their importance as land resources. (5 Marks)

- Plains: About 43 per cent of India's land is plain area, providing facilities for agriculture and industry .
- Mountains: Account for 30 per cent of total surface area, ensuring perennial flow of rivers and providing facilities for tourism and ecological aspects .
- Plateaus: About 27 per cent of the country is the plateau region, possessing rich reserves of minerals, fossil fuels, and forests .
- Careful planning is required to use this finite asset of land for various purposes .
- A pie chart showing land under important relief features (Plains 43

Q24: Explain the various categories of land utilisation in India. (5 Marks)

- Forests: Land dedicated to forest cover .
- Land not available for cultivation: Includes barren and waste land, and land put to non-agricultural uses like buildings and roads .
- Other uncultivated land: Includes permanent pastures, land under miscellaneous tree crops, and culturable waste land left for more than 5 years .

- Fallow lands: Current fallow (left for 1 year or less) and other than current fallow (left for 1 to 5 years) .
- Net Sown Area: The physical extent where crops are actually sown and harvested .
- A comparative diagram of land use categories between 1960-61 and 2019-20 would show marginal changes in forest and net sown area .

Q25: Explain how technology and colonisation played a role in resource exploitation. (5 Marks)

- History reveals that rich resources in colonies were the main attraction for foreign invaders .
- Higher levels of technological development in colonising countries helped them exploit resources of other regions .
- Colonising powers established supremacy by using technology to access and transform materials into resources .
- In India, resource development involves not just availability, but also technology, quality of human resources, and historical experience .
- Resources can contribute to development only when accompanied by appropriate technological and institutional changes .

Q26: Describe Alluvial soil in detail, covering its formation, distribution, and chemical properties. (5 Marks)

- It is the most widely spread soil in India, making up the entire northern plains .
- Formation: Deposited by three important Himalayan river systems—the Indus, the Ganga, and the Brahmaputra .
- Distribution: Found in the northern plains, Rajasthan, Gujarat, and eastern coastal plains (deltas of Mahanadi, Godavari, Krishna, and Kaveri) .
- Composition: Consists of various proportions of sand, silt, and clay .
- Chemical properties: Contains adequate proportions of potash, phosphoric acid, and lime .
- It is ideal for growing sugarcane, paddy, wheat, and cereal crops, leading to intensive cultivation and high population density .

Q27: Explain the characteristics and distribution of Black Soil and Laterite Soil. (5 Marks)

- Black Soil: Also known as regur or black cotton soil, it is made of extremely fine clayey material and has high moisture-holding capacity .
- Black Soil Distribution: Typical of the Deccan trap (Basalt) region; covers plateaus of Maharashtra, Saurashtra, Malwa, and Madhya Pradesh .
- Laterite Soil: Derived from Latin 'later' (brick), it develops in tropical climates with alternate wet and dry seasons .
- Laterite Characteristics: Result of intense leaching due to heavy rain; acidic ($pH < 6.0$) and generally deficient in plant nutrients .
- Laterite Distribution: Found in southern states, Western Ghats of Maharashtra, Odisha, West Bengal, and North-east .
- Laterite is useful for tea, coffee, and cashew nut after proper conservation .

Q28: What is Soil Erosion? Describe its various types and the role of human and natural forces. (5 Marks)

- Soil Erosion is the denudation of the soil cover and subsequent washing down .
- Natural forces like wind, glaciers, and running water lead to erosion .
- Human activities like deforestation, over-grazing, construction, and mining disturb the balance between soil formation and erosion .
- Gully Erosion: Running water cuts deep channels in clayey soils, making land unfit for cultivation (bad land) .
- Sheet Erosion: Water flows as a sheet over large areas down a slope, washing away the topsoil .
- Wind Erosion: Wind blows loose soil off flat or sloping land .
- A diagram of Gully Erosion would illustrate the deep channels formed by running water[cite: 375].

Q29: Detail the soil conservation methods used in different terrains of India. (5 Marks)

- Contour Ploughing: Ploughing along contour lines to decelerate the flow of water down slopes .

- Terrace Farming: Cutting steps on slopes to restrict erosion; well-developed in Western and Central Himalayas .
- Strip Cropping: Dividing large fields into strips and leaving strips of grass to grow between crops to break wind force .
- Shelter Belts: Planting lines of trees to create shelter; helps stabilise sand dunes in western India .
- Afforestation and proper grazing management are general methods to check degradation .
- Proper discharge and treatment of industrial effluents help prevent soil pollution in suburban areas .

Q30: Describe the Soil Profile and factors contributing to soil formation. (5 Marks)

- Soil formation is a living system that takes millions of years to form a few cm in depth .
- Factors of formation: Relief, parent rock (bed rock), climate, vegetation, other forms of life, and time .
- Natural forces: Change in temperature, actions of running water, wind, glaciers, and activities of decomposers .
- Soil Profile Layers: Topsoil (upper layer), Subsoil (weathered rocks/sand/silt), Substratum (weathered parent rock), and Unweathered parent bed rock .
- Chemical and organic changes (humus) are equally important in defining soil properties .
- Refer to a diagram of the 'Soil Profile' to see the vertical arrangement of these layers .